

|          |                                                                                                |           |
|----------|------------------------------------------------------------------------------------------------|-----------|
|          |                                                                                                |           |
| 7(a)(i)  | (amplifier) gain is very large/infinite                                                        | <b>B1</b> |
|          | for amplifier not to saturate, $V^+ = V^-$<br><b>or</b><br>feedback (loop) ensures $V^+ = V^-$ | <b>B1</b> |
|          | $V^+$ is at earth/0 V so $V^-$ is (almost) at earth/0 V                                        | <b>B1</b> |
| 7(a)(ii) | gain = $(-5200 / 800)$ <b>or</b> $(-5.2 / 0.80)$                                               | <b>C1</b> |
|          | = -6.5                                                                                         | <b>A1</b> |

| Question | Answer                                                                                           | Marks       |
|----------|--------------------------------------------------------------------------------------------------|-------------|
| 7(b)     | (at saturation,) $V_{\text{OUT}} = 5 \text{ V}$                                                  | <b>C1</b>   |
|          | p.d. across R = $5 - 2.3$<br>$= 2.7 \text{ (V)}$                                                 | <b>C1</b>   |
|          | resistance = $2.7 / (30 \times 10^{-3})$<br>$= 90 \Omega$                                        | <b>A1</b>   |
|          | <b>or</b>                                                                                        |             |
|          | $R_{\text{diode}} = 2.3 / 0.030 = 77 \Omega$<br>$R_{\text{total}} = 5.0 / 0.030 = 167 \Omega$    | <b>(C1)</b> |
|          | $R_{\text{resistor}} (= 167 - 77) = 90 \Omega$                                                   | <b>(A1)</b> |
|          | <b>or</b>                                                                                        |             |
|          | $R_{\text{diode}} = 2.3 / 0.030 = 77 \Omega$<br>$77 / (R_{\text{resistor}} + 77) \times 5 = 2.3$ | <b>(C1)</b> |
|          | $R_{\text{resistor}} = 90 \Omega$                                                                | <b>(A1)</b> |
|          | <b>or</b>                                                                                        |             |
|          | $R_{\text{diode}} = 2.3 / 0.030 = 77 \Omega$<br>$R_{\text{resistor}} = 77 \times (2.7 / 2.3)$    | <b>(C1)</b> |
|          | $R_{\text{resistor}} = 90 \Omega$                                                                | <b>(A1)</b> |